Exp. 8 Uploading Architecture Diagram (MVC & Microservices) in Azure DevOps Wiki

DEFINITION:

An Architecture Diagram represents the overall structure of a system, showing components, their relationships, and how they interact.
MVC (Model-View-Controller) and Microservices architectures are commonly used to design scalable and maintainable applications.

PURPOSE:

- To represent system architecture clearly
- To show separation of concerns using MVC
- To illustrate distributed system design using Microservices
- To understand component interaction and data flow

BASIC COMPONENTS (Notations):

1. Model (MVC)

- Represents data and business logic
- Handles database interactions

Example: User Model, Product Model

2. View (MVC)

- Represents the UI layer
- Displays data to users

Example: Web page, Mobile UI

3. Controller (MVC)

- Handles user input and requests
- Connects Model and View

Example: Login Controller

4. Microservices

• Independent services performing specific tasks

• Communicate via APIs

Example: Auth Service, Payment Service

5. API Gateway

• Entry point for all client requests

• Routes requests to appropriate services

6. Database

• Stores data for services

• Can be separate for each microservice

7. Communication

• Interaction between services

• REST APIs / HTTP / Messaging

HOW TO UPLOAD ARCHITECTURE DIAGRAM

AIM:

To upload an Architecture Diagram (MVC & Microservices) in the Wiki using Azure DevOps.

TOOLS REQUIRED:

• Azure DevOps

• Web browser

• Architecture Diagram image (PNG/JPG)

PROCEDURE:

Step 1: Open Azure DevOps

• Go to Azure DevOps

• Sign in with your Microsoft account

Step 2: Select Project

• Choose your existing project from the dashboard

Step 3: Navigate to Wiki

• Click on Wiki from the left menu

• If Wiki is not created:

o Click Create Wiki

o Select Project Wiki

Step 4: Create New Page

• Click on New Page

• Enter Architecture Diagram

Step 5: Upload Diagram

• Click on Insert Image (image icon)

• Choose Upload from Computer

• Select the Architecture Diagram file

• Click Upload

Step 6: Save the Page

• Click Save

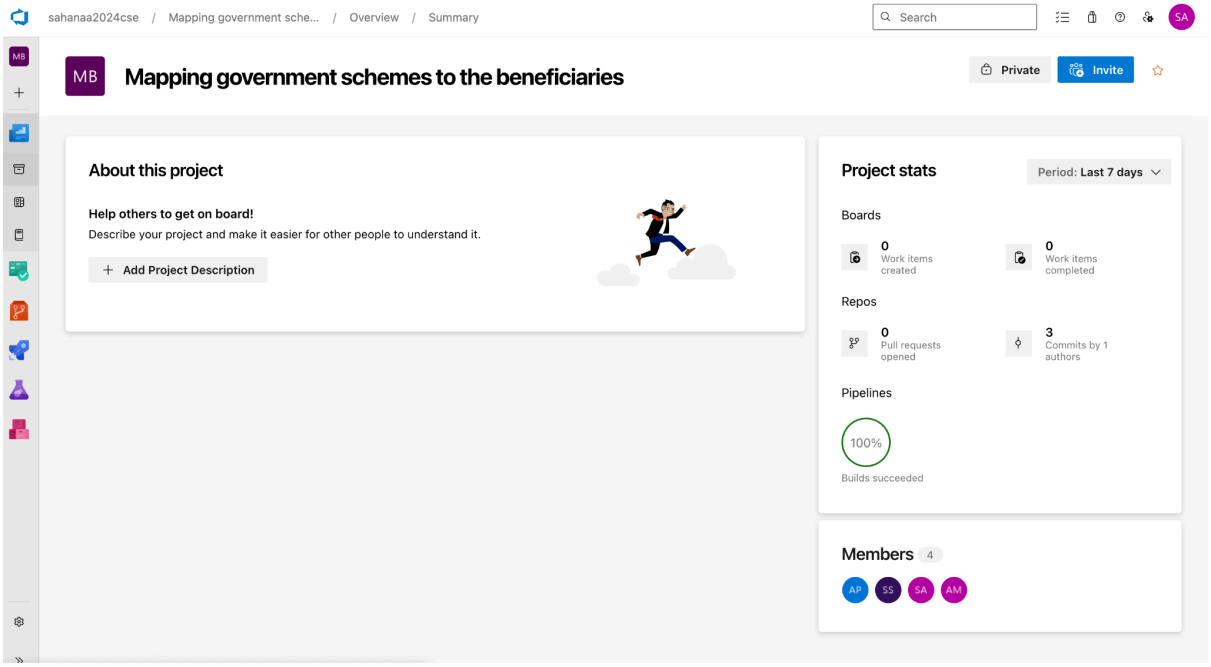
Step 7: Verify Upload

• Ensure the Architecture Diagram is displayed correctly

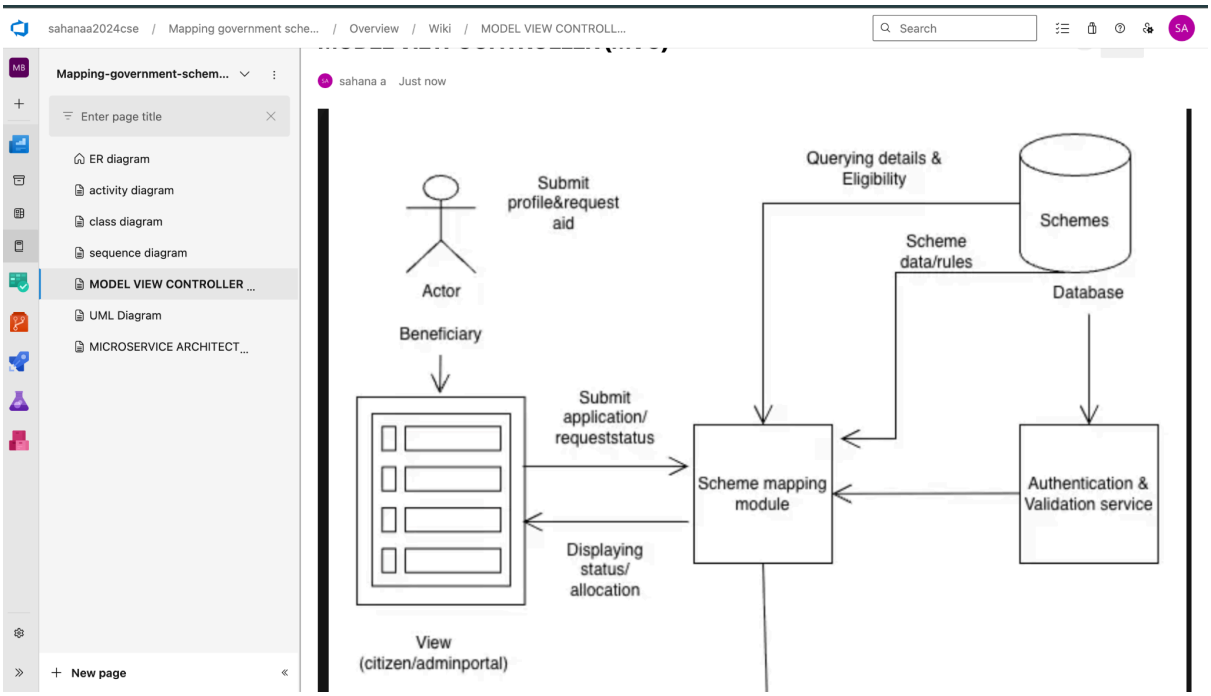
RESULT:

Thus, the Architecture Diagram using MVC and Microservices was successfully uploaded to the Wiki using Azure DevOps.

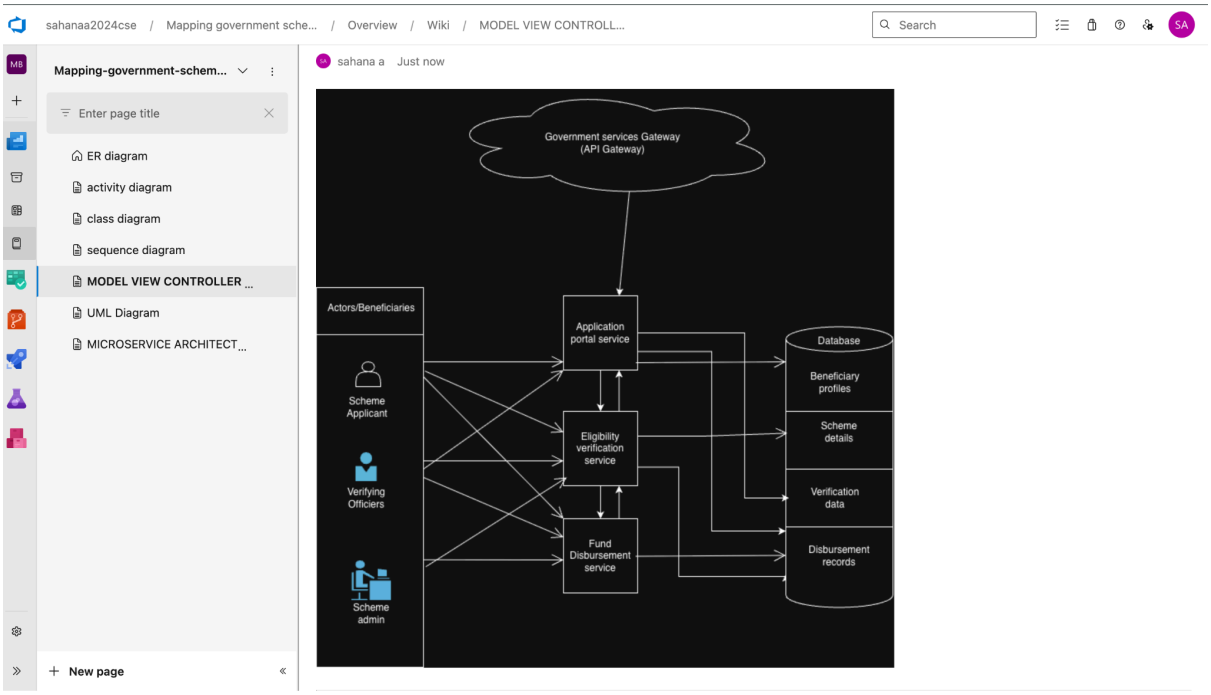
Project Page



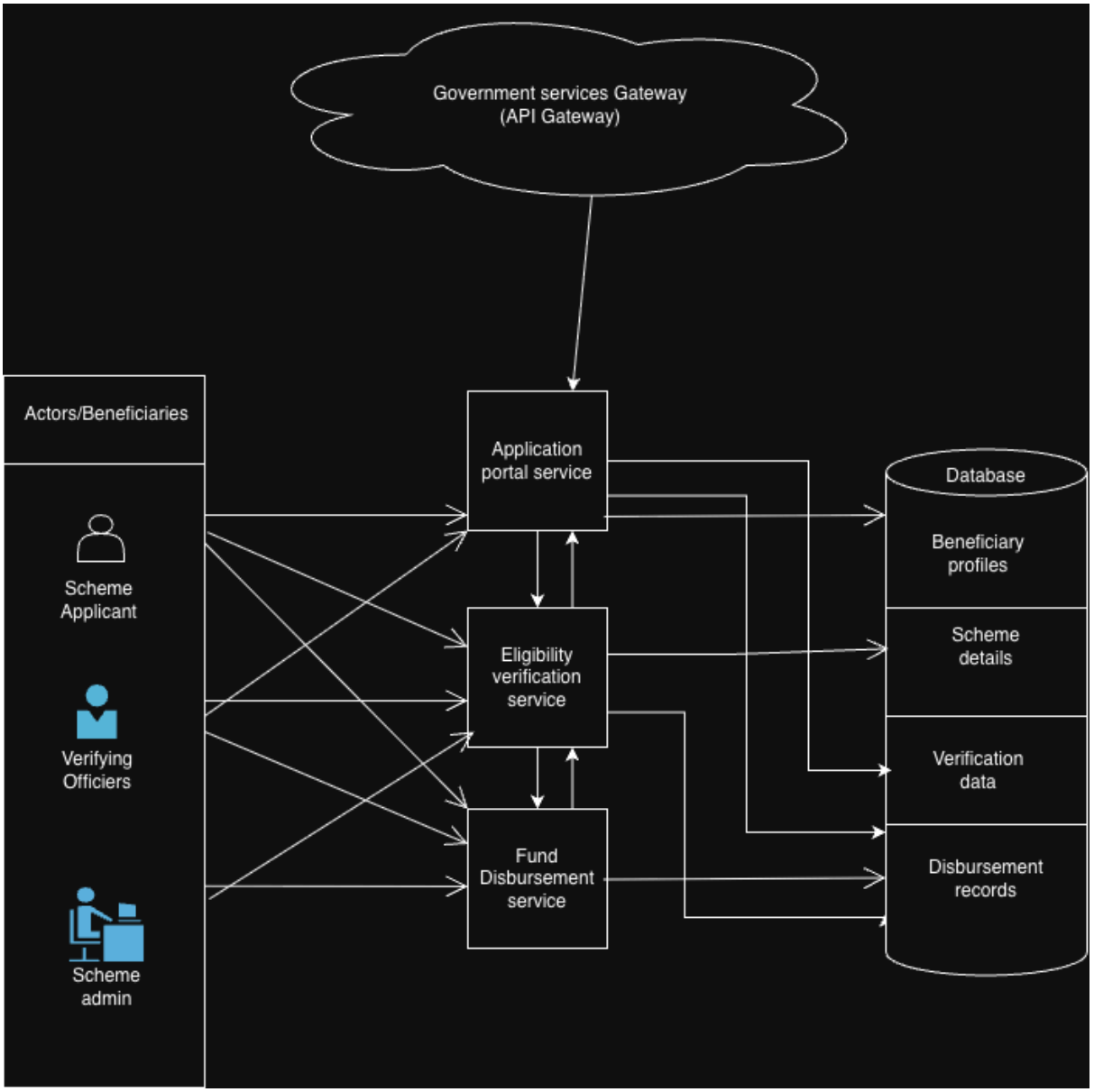
Wiki Page

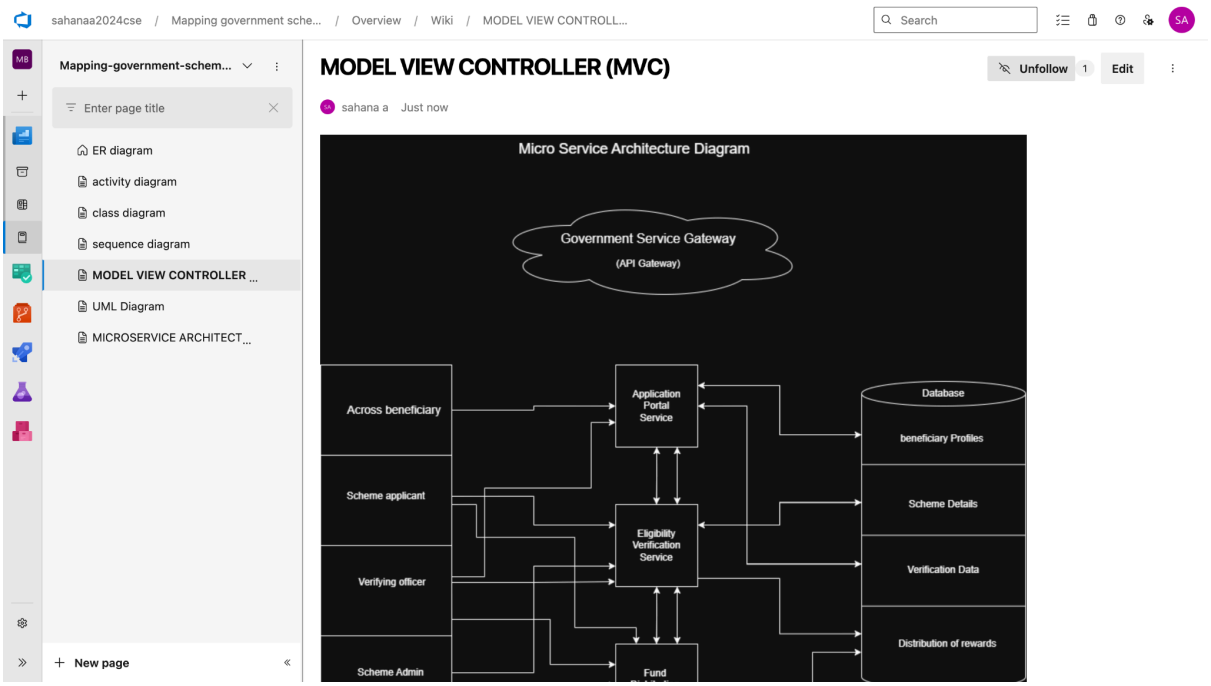


MVC Architecture Diagram



Microservice Architecture Diagram

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MICROSERVICE ARCHITECT...
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MODEL VIEW CONTROLLER (MVC)
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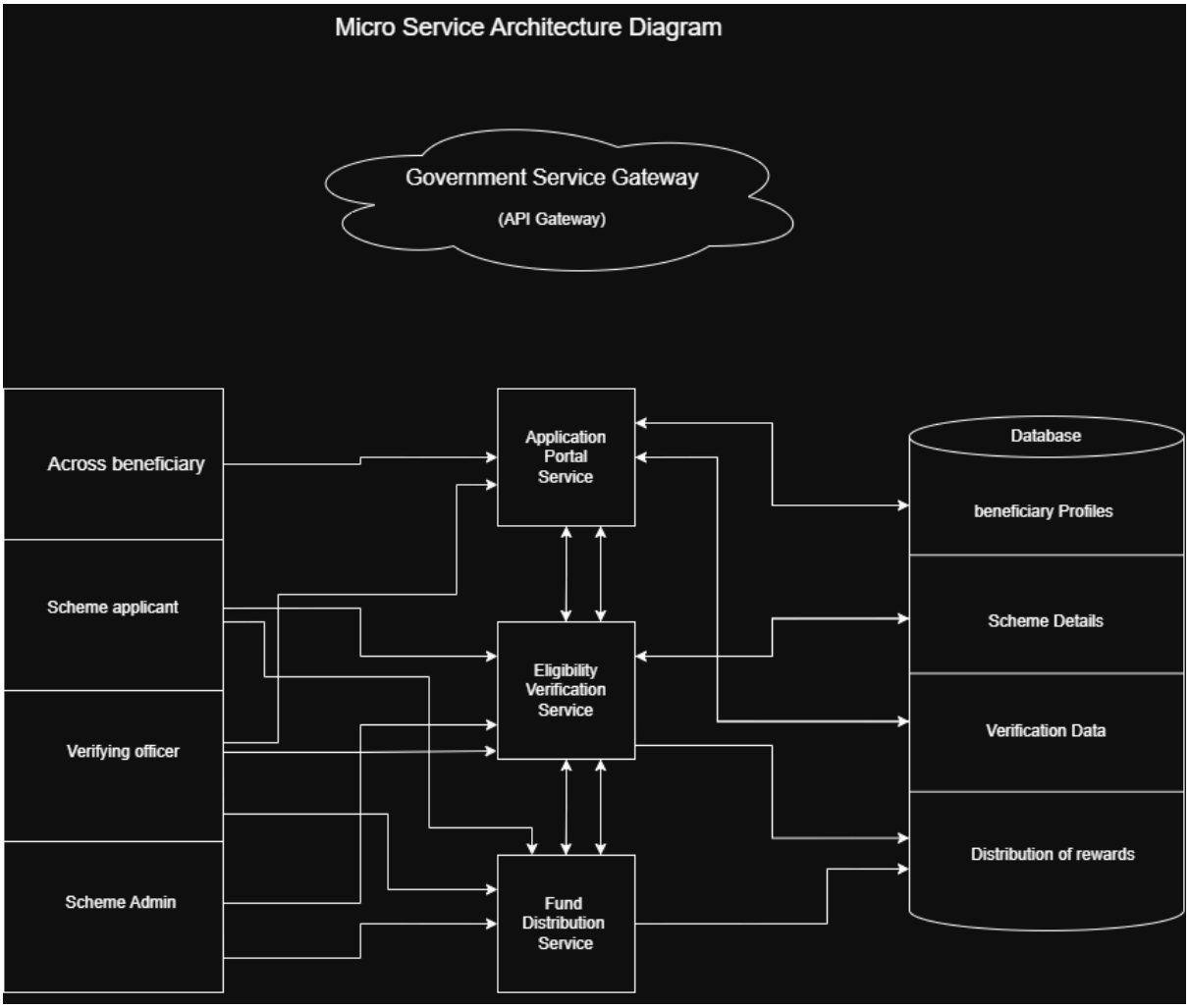
Micro Service Architecture Diagram
Government Service Gateway
(API Gateway)
Across beneficiary
Scheme applicant
Verifying officer
Scheme Admin
Application Portal Service
Eligibility Verification Service
Fund
Database
beneficiary Profiles
Scheme Details
Verification Data
Distribution of rewards


Micro Service Architecture Diagram
Government Service Gateway
(API Gateway)
Across beneficiary
Scheme applicant
Verifying officer
Scheme Admin
Application Portal Service
Eligibility Verification Service
Fund Distribution Service
Database
beneficiary Profiles
Scheme Details
Verification Data
Distribution of rewards